THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–14) | Class 8 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** 15	Squares from 15^2 to 29^2
2	**(a)** 2^8	$2 \times 2^7 = 2^8$
3	**(b)** 5	$3b + 4 = 19$
4	**(b)** 90°	One part $= 18^\circ$
5	**(d)** 41	LCM plus the common remainder
6	**(d)** 53	$x^2 + y^2 = (x+y)^2 - 2xy$
7	**(a)** 8 : 12 : 21	Make the middle terms equal using their LCM
8	**(b)** 1%	Loss % = (p²/100) = 10²/100
9	**(c)** 120	Legs 20 and 48 since $5^2 + 12^2 = 13^2$
10	**(d)** 24	Let ages be $3x$, $5x$: $(3x+8) : (5x+8) = 2 : 3$ gives $x = 8$
11	**(c)** 20	$E = 5F/2 = 30$; $V = 2 + E - F$
12	**(a)** 42	$5 \times 30 - 4 \times 27$
13	**(d)** 2	$9(a-b) = 9d$
14	**(d)** 768 cm²	$l + b = 64$, $l = 3b$: $b = 16$, $l = 48$
15	**(b)** 62	Next square is $36^2 = 1296$
16	**(b)** 4	$2x + 1 = 9$
17	**(c)** 5	Number of binary digits of the largest weight to be measured
18	**(c)** 60°	$x + 2x = 180$
19	**(b)** 57	71 multiples up to 500 minus 14 multiples below 100
20	**(c)** 49	$x^2 + y^2 = (x-y)^2 + 2xy$

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Next square 3969 (1); add 69, root 63 (1).	1+1
22	$243 = 3^5$ (1); $2x - 1 = 5$ gives $x = 3$ (1).	1+1
23	$2b + 3 = 13$ (1); $b = 5$ (1).	1+1
24	Opposite angles equal gives $x = 12$ (1); $\angle A = 62^\circ$, so $\angle B = 180 - 62 = 118^\circ$ (1).	1+1
25	LCM(20, 25) $= 100$ (1); number $= 100 + 3 = 103$ (1).	1+1
26	$x^2 + y^2 = (x+y)^2 - 2xy$ (1) $= 144 - 70 = 74$ (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Side $= \sqrt{1225} = 35$ m (1); perimeter $= 4 \times 35 = 140$ m (1); cost $= 140 \times 40 =$ ₹5600 (1).	1+1+1
28	Thickness doubles at each fold: 0.1×2^{10} (1); $= 102.4$ mm (1); in cm $= 10.24$ cm (1).	1+1+1
29	$45 = 101101_2$ (1); weights used: 1, 4, 8, 32 g (1); each 1 means the weight is used (1).	1+1+1
30	$\angle B = 130^\circ$ (1); $\angle PAB = 25^\circ$ and $\angle PBA = 65^\circ$ (1); $\angle APB = 180 - (25 + 65) = 90^\circ$ (1).	1+1+1
31	$247247 = 247 \times 1001$ (1); $1001 = 7 \times 11 \times 13$ (1); so divisible by 7, 11 and 13 (quotients 35321, 22477, 19019) (1).	1+1+1
	OR: $x^2 + y^2 = (x-y)^2 + 2xy = 16 + 42 = 58$ (1.5); $(x+y)^2 = (x-y)^2 + 4xy = 16 + 84 = 100$ (1.5).	1½+1½
32	Let ages be $3x$ and x (1); $(3x+4) \times 3 = (x+4) \times 7$ gives $x = 8$ (1); ages 24 and 8 years (1).	1+1+1
	OR: CP1 = 1980×100/110 = 1800 (1); CP2 = 1980×100/90 = 2200 (1); total CP − total SP = ₹ 40, a loss (1).	1+1+1

SECTION — D (Long Answer) — Answers — 6 × 5 = 30

Q.	Expected Answer / Value Points	Marks

33	LCM $= 24 = 2^3 \times 3$ (2); multiplier to make a perfect square 6 (2); required number 144 (1).	2+2+1
34	Number $= 5 \times 3^4$ (2); value 405 (2); it is $3^4 = 81$ times the original (1).	2+2+1
35	Digits are all less than b, so $b > 4$ (1); $2b^2 + 3b + 4 = 69$ (2); trying $b = 5$ gives $50 + 15 + 4 = 69$ (1); so $b = 5$ (1).	1+2+1+1
36	One part $= 360 \div 10 = 36°$ (1); angles $36°$, $72°$, $108°$, $144°$ (2); $\angle A + \angle D = 36 + 144 = 180°$ (1); these are co-interior angles, so AB $\parallel$ DC (1).	1+2+1+1
	OR: Splitting 45 into coprime factors and applying the rules (2); possibilities: a = 2, b = 0 (number 4230), a = 6, b = 5 (number 4635) (2); verification (1).	2+2+1
37	Area $= (3x + 2)(3x - 2) = 9x^2 - 4$ (2); at $x = 5$: 221 m^2 (1); square $= 9x^2$, difference $= 4$ m^2 (2).	2+1+2
	OR: Investment $\times$ time: 12, 12, 15 (1); ratio $4 : 4 : 5$ (1); one part ₹ 100 (1); shares ₹ 400, ₹ 400, ₹ 500 (2).	1+1+1+2
38	Year 1: A1 = ₹ 12000, interest ₹ 2000 (1); year 2: A2 = ₹ 14400, interest ₹ 2400 (2); year 3: A3 = ₹ 17280, interest ₹ 2880 (1); total ₹ 7280 = A3 − P (1).	1+2+1+1
	OR: $AC^2 = 9^2 + 12^2 = 225$, $AC = 15$ cm (1); $AC^2 + CD^2 = 225 + 400 = 625 = DA^2$ so $\angle ACD = 90°$ (2); area $= \frac{1}{2} \times 9 \times 12 + \frac{1}{2} \times 15 \times 20 = 204$ cm² (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–14) | Class 8 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** 24 years	Let the ages be $3x$ and $2x$; solving $(3x+4)\times 5=(2x+4)\times 7$ gives $x=8$
2	**(c)** 60500	50000×(1+10/100)²
3	**(a)** 972	Sides 27 and 36; area $=27\times 36$
4	**(b)** 1350	One part $=$ ₹450
5	**(d)** 8	$V=2E/3=12$; $F=2+E-V$
6	**(b)** 41	Correct sum $=360-30+80$
7	**(b)** 7	$(x+1/x)^2=x^2+2+1/x^2$
8	**(b)** 128 cm²	Height $=\sqrt{10^2-6^2}=8$ cm; area $=\frac{1}{2}(10+22)\times 8$
9	**(d)** 77	Previous square is $50^2=2500$
10	**(c)** 7	$2^4\times 5^7=10^4\times 125$
11	**(d)** 32_5	Convert every number to decimal and compare
12	**(b)** 130°	A diagonal bisects the vertex angles; $\angle BCD=2x$ and $\angle B$ is supplementary to it
13	**(a)** 112	Count multiples from 100 to 999
14	**(d)** 24000	$(512+488)(512-488)$
15	**(c)** ₹ 600	Convert the reciprocals to whole-number ratio using the LCM
16	**(c)** 40500	50000×(1−10/100)²
17	**(b)** 15, 112, 113	Use $(n,\frac{n^2-1}{2},\frac{n^2+1}{2})$ for odd n and $(n,\frac{n^2}{4}-1,\frac{n^2}{4}+1)$ for even n
18	**(b)** 104	$13\times 360/45$
19	**(a)** 24	Each corner is replaced by 3 new vertices: $8\times 3=24$
20	**(b)** 43	$(29+1)(14)-29\times 13$

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Perimeters $3:7$ (1); areas $9:49$ (1).	1+1
22	Final = (1+25/100)(1−10/100) (1) = 112.5% of original, so more by 12.5% (1).	1+1
23	Let legs be $3x$, $4x$; hypotenuse $=5x=35$ (1); $x=7$, legs 21 cm and 28 cm (1).	1+1
24	Take $x=3k$, $y=4k$ (1); ratio $=23k:8k=23:8$ (1).	1+1
25	$12(n-2)=48$ gives $n=6$ (1); total $=6^3=216$ (1).	1+1
26	Total $=6\times 20=120$ (1); rest $=5\times 18=90$, removed $=30$ (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Foot to first wall $=\sqrt{37^2-35^2}=12$ m (1); foot to second wall $=\sqrt{37^2-12^2}=35$ m (1); width $=47$ m (1).	1+1+1
28	Remaining provisions $=(45-15)\times 240=7200$ man-days (1); men left $=180$ (1); days $=40$ (1).	1+1+1
29	$E=3F/2=48$ (1); $V=2+E-F=18$ (1); $F+V-E=32+18-48=2$ (1).	1+1+1
30	New sum $=2480-38+80=2522$ (2); mean $=63.05$ (1).	2+1
31	$11(a+b)=110$ so $a+b=10$ (1); $a-b=4$ gives $a=7$, $b=3$ (1); number $=73$ (1).	1+1+1
	OR: $b=20$ m, $l=60$ m (1); area $=1200$ m² (1); fencing cost $=160\times 15=$ ₹ 2400 (1).	1+1+1
32	Number $=6^3=216$ (1); corner cubes have three painted faces (1); there are 8 of them (1).	1+1+1
	OR: 2 min $=120$ s (1); distance $=3\times 10^8\times 120$ (1); $=3.6\times 10^{10}$ m (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Let ages be $5x$, $8x$ (1); $(5x+6):(8x+6)=2:3$ gives $x=6$ (2); ages 30 and 48 years (1); 6 years ago: 24 and 42, ratio $4:7$ (1).	1+2+1+1
34	$F=15$; $E=(3\times10+4\times5)/2=25$ (2); $V=2+E-F=12$ (2); $F+V-E=2$ (1).	2+2+1
35	Total $=15\times65+25\times15+10\times25=1600$ (3); mean $=32$ (2).	3+2
36	Using $(p+q)^2-(p-q)^2=4pq$ (2); $4\times2n\times3=24n$ (1); for $n=4$: 96 (2).	2+1+2
	OR: Other diagonal $=2\times216/18=24$ cm (2); side $=\sqrt{9^2+12^2}=15$ cm (1); altitude $=216/15=14.4$ cm (2).	2+1+2
37	LCM $=36=2^2\times3^2$ (2); multiplier to make a perfect cube 6 (2); required number 216 (1).	2+2+1
	OR: A is heavier (1); ratio $=\frac{6}{3}\times10^{24-22}=200$ times (2); total 6.03×10^{24} kg (2).	1+2+2
38	Decimal values 187 and 96 (1); difference 91 (2); 331_5 (1); check by converting back (1).	1+2+1+1
	OR: Opposite sides are equal: $3x+4=2x+9$ gives $x=5$ (2); AB = CD $=19$ cm (1); perimeter $=2(19+12)=62$ cm (2).	2+1+2

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–14) | Class 8 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** 7	$4d + 16$ = sum
2	**(d)** 1080	$1000 \times 120/100 \times 90/100$
3	**(d)** 29 and 30	$(n+1)^2 - n^2 = 2n + 1$ gives $n = 29$
4	**(c)** 3	Ones digits repeat in the cycle [7, 9, 3, 1]
5	**(b)** 4	$77 = 1001101_2$
6	**(a)** 50°	$\angle AOB = 180 - x$; OA = OB, so $\angle OAB = x/2$
7	**(a)** 111	Test palindromes 101, 111, 121, ...
8	**(d)** 7	Square both sides
9	**(b)** 9 days	Find the work left in man-days and divide by the new number of men
10	**(b)** −4%	(1+20/100)(1−20/100) − 1
11	**(b)** 5280	Side 73; half other diagonal 55; area $= \frac{1}{2}d_1d_2 = 5280$
12	**(b)** 140	Combine to $a : b : c$ first
13	**(b)** 4	$6(n-2)^2 = 24$ gives $n = 4$
14	**(d)** 12	Sum of first n odd numbers is n^2, so mean $= n$
15	**(d)** 18	$a^2 - b^2 = (a-b)(a+b)$
16	**(d)** 15	$2A/16$
17	**(d)** 3	A k-digit square has $\lceil k/2 \rceil$ digits in its root
18	**(c)** 27	$x = 5$, so 3^3
19	**(d)** MCMXCIV	Break 1994 into thousands, hundreds, tens and units
20	**(c)** 110°	$\angle A + \angle D = 180^\circ$ gives x; base angles $\angle C = \angle D$

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	$a + b = 9$, $a - b = 3$ (1); $a = 6$, $b = 3$, number 63 (1).	1+1
22	$l + b = 50$, $l - b = 18$ (1); $l = 34$, $b = 16$, area $= 544$ cm² (1).	1+1
23	$(n+1)^2 - n^2 = 2n + 1$ (1) so $n = 37$; numbers 37 and 38 (1).	1+1
24	Convert to the same power or usual form: 340000 (1); 3.4×10^5 (1).	1+1
25	$90 = 2 \times 45 + 0$; $45 = 2 \times 22 + 1$; $22 = 2 \times 11 + 0$; $11 = 2 \times 5 + 1$; $5 = 2 \times 2 + 1$; $2 = 2 \times 1 + 0$; $1 = 2 \times 0 + 1$ (1); 1011010_2 has 4 ones (1).	1+1
26	One part $= 360 \div 20 = 18^\circ$ (1); largest $= 126^\circ$, which is obtuse (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Decimal values 98 and 57 (1); sum 155 (1); 1110_5 (1).	1+1+1
28	$10x - 60 = 360$ (1); $x = 42$ (1); angles 42°, 74°, 106°, 138° (1).	1+1+1
29	LCM $= 24$ (1); 9999 divided by it leaves remainder 15 (1); number 9984 (1).	1+1+1
30	$(3x+2)^2 = 9x^2 + 12x + 4$ (1); $(3x-2)^2 = 9x^2 - 12x + 4$ (1); difference $= 24x$ (1).	1+1+1
31	Total work $= 360$ man-days; done $= 120$ (1); left $= 240$ (1); days $= 240 \div 20 = 12$ (1).	1+1+1
	OR: V = P(1−r/100)³ (1) = 500000×90/100 three times (1) = ₹ 364500 (1).	1+1+1
32	Side $= 73$ cm; half-diagonal $= 48$ cm (1); other half $= \sqrt{73^2 - 48^2} = 55$ cm, other diagonal 110 cm (1); area $= \frac{1}{2} \times 96 \times 110 = 5280$ cm² (1).	1+1+1
	OR: Let milk $= 3k$, water $= 2k$ (1); $(2k+10)/(3k) = 4/3$ gives $k = 5$ (1); milk $= 15$ L, water $= 10$ L (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	832-238=594 (1); reverse of 594 is 495 (1); 594+495=1089 (1); in general the difference is $99(a-c)$, whose digits are $(a-c-1), 9, (10-a+c)$ and the sum is always 1089 (2).	1+1+1+2
34	$xy = \dfrac{(13)^2 - 97}{2} = 36$ (2); $(x-y)^2 = 97 - 2(36) = 25$ (2); $x - y = 5$ (1).	2+2+1
35	Milk 40 L, water 20 L (1); in 15 L removed: milk 10 L, water 5 L (1); milk left 30 L (1); water now $20 - 5 + 15 = 30$ L (1); ratio $1:1$ (1).	1+1+1+1+1
36	Let CP = x. SP = x(1+50/100)(1−20/100) = 1.2x (2); profit = 0.2x = 240 (2); x = ₹ 1200 (1).	2+2+1
	OR: (i) $\sqrt{65^2 - 60^2} = 25$ m (2); (ii) new height 56 m, $\sqrt{65^2 - 56^2} = 33$ m (2); (iii) $33 - 25 = 8$ m (1).	2+2+1
37	Work left after 10 days $= (30-10) \times 24 = 480$ man-days (2); workers now $= 16$ (1); more days $= 30$ (1); total $= 40$ days, delay $= 10$ days (1).	2+1+1+1
	OR: Stage 0 perimeter $= 81$ cm (1); each stage multiplies the perimeter by $\frac{4}{3}$ (1); stage 3: $3 \times 27 \times (4/3)^3 = 192$ cm (2); increase $= 111$ cm (1).	1+1+2+1
38	New mean $= 15 \times 4 - 5 = 55$ (3); new sum $= 12 \times 55 = 660$ (2).	3+2
	OR: $(100a + 10b + c) - (100c + 10b + a) = 99a - 99c$ (2); $= 99(a-c) = 99 \times 1$ (1); difference $= 99$ (2).	2+1+2

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